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Research on Dynamic Sign Language Algorithm Based on Sign Language Trajectory and Key Frame Extraction

Yufei Yan
School of Automation
Wuhan University of Technology
Wuhan 430070, China
e-mail: 877018321@qq.com

Zhijun Li*
School of Automation
Wuhan University of Technology
Wuhan 430070, China
e-mail: 460254218@qq.com

Qunzhu Tao,
School of Automation
Wuhan University of Technology
Wuhan 430070, China
e-mail: taoqunzhu@163.com

Chenyu Liu
School of Automation
Wuhan University of Technology
Wuhan 430070, China
e-mail: 2442338480@qq.com

Rui Zhang
School of Automation
Wuhan University of Technology
Wuhan 430070, China
e-mail: 1025427063@qq.com

Abstract—**Based on the depth information of Kinect, this paper studies the real-time dynamic sign language recognition algorithm and improves the dynamic time warping algorithm for the recognition of sign language trajectories. The traditional DTW algorithm is improved by using the path constraint of the relaxed endpoint, adding the lower bound function to cull part of the candidate sequence and terminate the match early. The key frames are extracted according to the sign language trajectory density, and the dynamic hand sign trajectory and key gesture type information are combined to obtain the final dynamic sign language recognition result. Experiments show that the average recognition rate of this method is more than 90%, which is better than the traditional DTW algorithm in recognition speed and recognition accuracy.**

Keywords-dynamic sign language recognition; Kinect sensor; DTW algorithm; key frame; sign language track

I. INTRODUCTION

With the rapid development of artificial intelligence and other technologies, gesture recognition as a major human-computer interaction method has gradually become a hot issue. As a special gesture, sign language is also the main communication method for language-disabled people. It carries a wealth of information and has a good expressive ability. Because people with no language barriers generally do not have sign language skills, the study of sign language recognition based on computer vision can not only facilitate the communication between language disabled and non-language obstacles, but also has important significance for the development of human-computer interaction.

For the recognition of dynamic gestures, the positional features and shape features of the dynamic gestures in continuous time are usually extracted. Commonly used dynamic gesture recognition methods are classified according to principles, such as statistics based (conditional random field and finite state machine), grammar based (hidden Markov model and dynamic Bayesian), and template-based (template matching and dynamic time warping) [1] methods. The DTW (dynamic time warping method) is selected in this paper. The Kinect sensor is used to extract the motion trajectory and key gesture in the sign language process as the basis of real-time sign language recognition. In addition, the DTW-based hand trajectory recognition strategy is studied and improved.

II. DTW AND ITS IMPROVEMENT

A. Hand Segmentation and Tracking Method

Kinect2.0 can capture the depth information of the image. Segmentation of images using depth information can overcome the effects of traditional hand segmentation in unstable environments such as changes in illumination or complex background changes. This paper uses a hand segmentation method that combines depth threshold and skin color threshold.

The TLD (Tracking-Learning-Detection) algorithm is selected as the method of hand tracking in this paper. It is an

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algorithm that works well for occlusion, deformation, etc. [2]. Moreover, the TLD algorithm continuously updates the "significant feature points" of the tracking module and the target model of the detection module and related parameters through an improved online learning mechanism, thereby making the tracking more stable, robust and reliable.

B. The Basic Principle of the DTW Algorithm

The DTW algorithm transforms the global optimal problem into a local optimal problem by scaling and distorting the time series of the sample. It looks for an optimal path between the two samples, the sum of the distances of the paths as the distance between the two time series.

For the two unequal lengths of time series $T=\{t_1,t_2,\cdots t_m\}$ and $R=\{r_1,r_2,\cdots r_n\}$, find the best path between them. According to the length of the sequences T and R, a matrix P of $n\times m$ is constructed:

$$P(i,j)=\delta(t_i,r_j) \tag{1}$$

δ represents the metric distance between two points, generally chosen as the Euclidean distance. $(0,0)$ of the mark matrix P is the start point of the Warp Path W, and (n,m) is the end point. W can be expressed as:

$$W=w_1,w_2,\cdots w_l \qquad (\max(n,m)\le l\le m+n) \tag{2}$$

where in $w_k=(i,j)$ represents the path of the k-th element. The path W can only be moved successively to the right or upper point adjacent to the previous point. DTW constructs a cumulative cost (DTW distance) matrix D by the principle of dynamic programming, and $D(i,j)$ represents the DTW distance [3] of the path W from the starting point to the point (i,j). The formula is

$$D(i,j)=P(i,j)+\min\begin{cases}D(i-1,j)\\D(i-1,j-1)\\D(i,j-1)\end{cases} \tag{3}$$

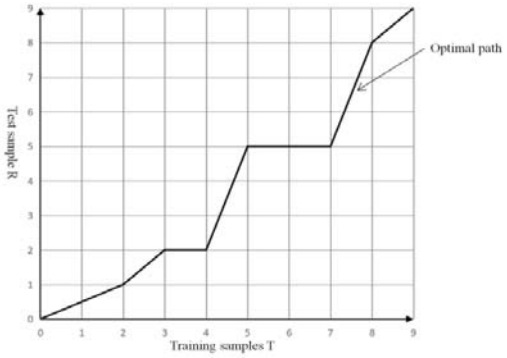


Figure 1. DTW optimal path diagram.

Based on the original DTW algorithm, the DTW algorithm is optimized by conditional constraints, lower bound distance filtering and early termination matching.

C. Boundary Constraint with Relaxed Endpoint

In the actual operation of the DTW, it will be found that if the search range is too large, there are many areas where no path branches. Therefore, constraining the search boundary can remove the invalid area, thereby reducing the amount of calculation and improving the calculation efficiency.

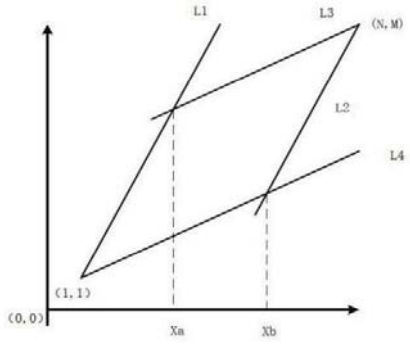


Figure 2. Select the search boundaries.

If the search area is too large, the limited area will have less effect. However, if the area is too small, it will have a greater impact on the recognition rate. It has been found through experiments that the boundary effect is best when the slope is between 0.5 and 1.

Delimit the constraint interval, then according to the coordinates (N, M) can be calculated:

$X_a=(2M-N+2)/3$, $X_b=(4N-2M+1)/3$

When the search path is in this area, it satisfies:

$$\begin{cases}2M-N\ge 3\\2N-M\ge 2\end{cases} \tag{4}$$

In the process of sign language recognition, the obtained sign language trajectory may capture the redundant trajectory information when not in the sign language, which has a certain influence on the gesture recognition. Therefore, the error caused by redundant information is eliminated by relaxing the matching endpoints.

he specific method of relaxing the endpoint is as follows: In order to match the first frame and the last frame of the sample to be detected to the first L frame and the last L frame of the sample to be matched, respectively, the previously constrained diamond path is respectively relaxed to both sides. As shown in Fig 3, the starting area of the search may be a point on (1, 1) to (1, L) or (1, 1) to (L, 1). The end point of the search path may be (N-L+1, M) to (N, M) or a point on (N, M-L+1) to (N, M).

According to the relaxed area, get a hexagon. Analysis of the hexagonal region yields values for X_a and X_b :

$X_a=(2M-N-L)/3$, $X_b=2\times(2N-M+L/2)/3$

Round up X_a and X_b to get the relationship between M and N as follows:

$$\begin{cases} 2M - N \geq 3 + L \\ 2N - M \geq L \end{cases} \tag{5}$$

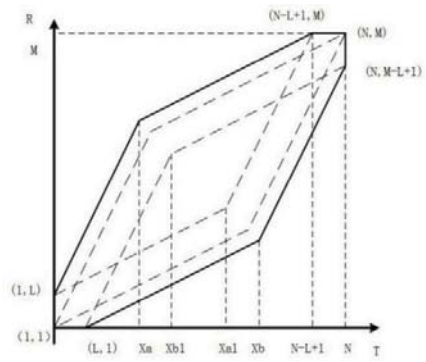


Figure 3. Schematic diagram of endpoint relaxation path constraints.

D. Lower Bound Distance Filtering of DTW

In the process of performing DTW matching, the samples to be tested need to be compared with the template samples one by one. A lower bound function is used to filter out a part of the template sequence, and then the comparison calculation is performed according to the DTW algorithm. The time complexity calculated by this method is lower than the direct calculation of the DTW distance, which can greatly improve the matching efficiency [4]. The necessary condition for filtering using the lower bound function is that the calculated value of the lower bound function for the two time series must be smaller than the DTW distance between them, and the result of the metric is as close as possible to the measurement result of the DTW algorithm [5].

In [6], Keogh et al. proposed the LB_Keogh lower bound function and verified that its effect is better than the LB_Yi and LB_Kim lower bound functions. The idea of the LB_Keogh lower bound function is to first calculate the envelope of a sample sequence to be tested, and the sample to be tested is between the upper envelope and the lower envelope. Then, the template sample sequence and the envelope are calculated, and the partial size of the template sample sequence outside the envelope range is taken as the final value of the lower bound function. But LB_Keogh can only filter two sequences of equal length. In the process of actually playing sign language, it is difficult to guarantee that the same sign language is a time series of equal length each time.

This paper proposes a lower bound function of LB_BC classified according to the relative position of the sequence, and the lower bound function is given according to the relative positions (intersection, separation, inclusion) of the two time series. Suppose there are two time series $Q\langle q_1, q_2, \cdots q_n \rangle$ and $C\langle c_1, c_2, \cdots c_m \rangle$, and $\max(q_1, q_2, \cdots q_n) > \max(c_1, c_2, \cdots c_m)$.

Then when Q and C are completely separated, $\min(q_1, q_2, \cdots q_n) \geq \max(c_1, c_2, \cdots c_m)$, denoted as case a. As shown in Fig 4(a).At this time, the lower bound function D_{LB_BC} is:

$$D_{LB_BC} = \max\left\{\sum_{i=1}^{n} f(q_i - \max(c_1, c_2, \cdots c_m)), \sum_{j=1}^{m} f(\min(q_1, q_2, \cdots q_n) - c_j)\right\} \tag{6}$$

When Q and C are not completely separated and are not contained, $\min(q_1, q_2, \cdots q_n) \leq \max(c_1, c_2, \cdots c_m)$ and $\min(q_1, q_2, \cdots q_n) > \min(c_1, c_2, \cdots c_m)$, denoted as case b. As shown in Fig 4(b).At this time, the lower bound function D_{LB_BC} is:

$$D_{LB_BC} = \sum_{i=1}^{n} f(q_i, \max(c_1, c_2, \cdots c_m)) + \sum_{j=1}^{m} f(\min(q_1, q_2, \cdots q_n), c_j) \tag{7}$$

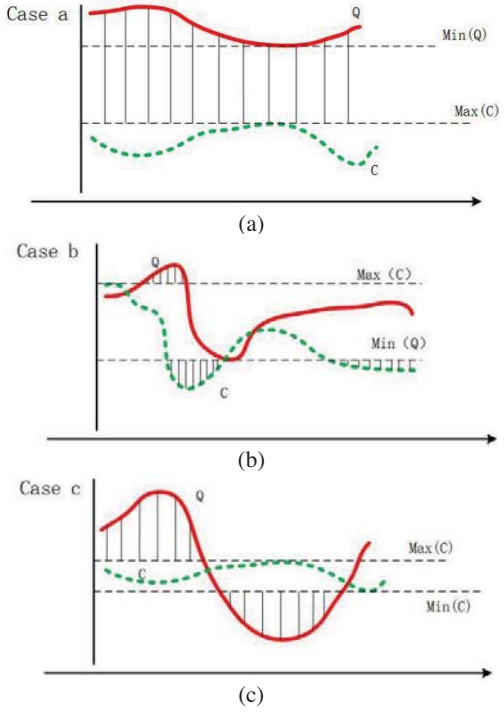


Figure 4. Schematic diagram of the lower bound function of LB_BC.

When Q and C are not completely separated and containable, $\min(q_1, q_2, \cdots q_n) \leq \max(c_1, c_2, \cdots c_m)$ and $\min(q_1, q_2, \cdots q_n) \leq \min(c_1, c_2, \cdots c_m)$, denoted as case c .As shown in Fig 4(c).At this time, the lower bound function D_{LB_BC} is:

$$D_{LB_BC} = \sum_{i=1}^{n} f(q_i, \max(c_1, c_2, \cdots c_m)) + \sum_{j=1}^{m} f(\min(q_1, q_2, \cdots q_n), q_i) \tag{8}$$

Then, the classification-based lower bound function D_{LB_BC} can be constructed according to Equations 6, 7, and 8. In Equations 6, 7, 8, the function:

$$f(x,y)=\begin{cases}|x-y| & x>y\\ 0 & other\end{cases} \tag{9}$$

According to Equations 6, 7, and 8, the calculated time complexity is M+N, so using the LB_BC lower bound function has a computational complexity lower than DTW. For the calculation of the lower bound function of the two time series, the early termination matching strategy can still be used, which will not be introduced here.

III. Sign Language Recognition Strategy Combining Motion Trajectory and Key Gesture

A. Joint Point Weight Assignment of Hand Trajectory

In the process of sign language recognition, different joint points can bring different information. For the two-hand sign language, the amount of information brought by the two palms at different times is different [7]. The weight of the joint point can be given according to the magnitude of the motion of the hands in the different sign language of the sample to be tested.

Calculate the total displacement of a hand before the sign language trajectory preprocessing:

$$D_i=\sum_{t=2}^{T} Dist^i\left(p_t,p_{t-1}\right) \tag{10}$$

where Pt represents the position of the t-th frame of the i-th skeletal point motion trajectory.

Then the trajectory sequence for the skeletal point i is assigned a weight in the DTW match: $\omega_i=D_i\Big/\sum_{j=1}^{N}D_j$.

Then the DTW distance between the test sample and the template sample:

$$d(m_k,n_k)=\sum_{i=1}^{N} Dis^i\left(p_{m_k},p_{n_k}\right)\omega_i \tag{11}$$

where $Dis^i\left(p_{m_k},p_{n_k}\right)$represents the DTW distance between the test template and the sample template of the i-th skeletal point.

B. Key Gesture Extraction Based on Sliding Window

When extracting the static hand information in the dynamic sign language, the gesture expressed by the sign language of many frames are useless transition frames, so the gesture analysis and feature extraction of the key frames are needed to improve the real-time performance of the system. The key frame of the video is extracted using the frame difference method [8]. The frame difference method first uses the first frame as a key frame, and compares the first frame with all other frames by setting a threshold to remove frames smaller than the threshold. Then, the frame that has not been removed is used as the initial frame, and so on until the distance between all the frames is large enough.

For sign language of both hands, each frame image acquired by Kinect contains the x, y, and z-axis coordinates of the two track points of the two hands. It is also a 6-dimensional vector. Calculating the intensity of a complete sign language requires calculating the density of the left and right hand coordinates separately and then adding them.

The density of a point P is the size of all the point sets in which the track T at which the point P is located is within a fixed threshold ε . Define point collections:

$$P_X=\left\{X \mid \forall X\in T,\delta(P,X)<\varepsilon\right\} \tag{12}$$

$\delta(P,X)$ is the Euclidean distance of point P and point X, and ε is the threshold. The points P are traversed forward and backward on the time series, respectively, until the traversed points do not belong to the set P_X , and then the points in the set P_X that are not traversed are removed. The size of the obtained point set P_X' is the density of the point P in the time series:

$$C(P)=\left\|P_X'\right\| \tag{13}$$

After the sign language density curve is obtained, the key frame is selected. The initial key frame is extracted from the frame corresponding to the extreme point of the density curve. After removing the start gesture and the termination gesture, in general, a sign language can contain up to 5 key gestures, and it is expected that there are enough differences between different gestures. This way each gesture can be representative.

Not the extreme points of the density values of all points are used as the initial key frame. First define a point density threshold σ and filter out points where the point density is less than the threshold σ . Select the average of the point density maximum and minimum as the threshold σ :

$$\sigma=\frac{\max\{C(P)\}+\min\{C(P)\}}{2} \tag{14}$$

After getting the valid part of the sign language word density distribution, key frames can be extracted. Directly extracting all the extreme points in the density function above the threshold σ will result in a redundant initial effective frame. If the initial effective frame is directly extracted by the frame difference method, the amount of calculation will increase. In this case, a sliding window based method is used to extract the initial key frames.

Assuming that the effective frame of the curve density is n, then the length of the sliding window is selected as $n/9$, when the number of frames on the left side of the window is n_s , and the number of frames on the right side is n_e ,

$n_e - n_s = n/9$ .The window moves from the left side to the right side. First, the extreme value point larger than the threshold value when $n_s = 0$ is selected as the key frame, and the left side of the window is moved to the position of the key frame after the selection is completed. If the initial window does not have a point that satisfies the condition, move the window to the right until an extreme point appears in the window. If the window has larger extreme points than the current key frame (the extreme point corresponding to the left window), discard the key frame on the left side of the current window and re-mark the larger extreme point as the key frame. Move the left side of the window to this point. This loops until there are no more extreme points in the window. Then move the window to the right until the right edge of the window reaches the next extreme point, then set the extreme point to the key frame and move the left side of the window to this point. Repeat these operations until $n_e > n$ .In this way, it is guaranteed that repeated key frames do not occur within $n/9$ consecutive frames.

After the key frame is obtained, the frame difference method is used to remove the redundant key gesture. The final result is the final key gesture of the dynamic sign language.

C. Identification Strategy

The DTW algorithm is used to identify the sign language trajectory, which can distinguish the sign language with a large difference in motion trajectory. However, the motion trajectory of much sign language is not very different, but there are some differences in the gesture in the process of demonstrating sign language, and the gesture of the same sign language may be different for different people or different situations. Therefore, sign language recognition is adopted by using the sign language trajectory as the main recognition basis and the key frame matching as the auxiliary basis.

For the pre-processed gesture trajectory, match the template sequence with the improved DTW algorithm. A threshold value σ is set, and it is assumed that the DTW distances of the m template tracks $T_1, T_2, \cdots, T_m$ and T which are closest to the trajectory of the sign language T to be tested are $d_1, d_2, \cdots, d_m$ in descending order. When $d_1/d_2 > \sigma$ is satisfied, it is considered that the template trajectory T_1 closest to the trajectory T to be measured is sufficiently similar to the trajectory T to be measured, and it can be determined that T belongs to the T_1 class. When $d_1/d_2 \le \sigma$, it is considered that T_1, T_2 , etc. may be the kind of track T to be tested, so that it is necessary to match the T and m template trajectories by gesture. The value of A determines the key degree of sign language trajectory and key gesture in the sign language recognition process to a certain extent. The larger the value of A, the lower the probability of success relying solely on trajectory recognition, and the more dependent on the recognition result of the gesture. Generally take $\sigma \in [1.2,2]$. The selection of m is positively correlated with the total number of template samples, generally taking $m \in [5,10]$.

The eigenvalue of each key gesture is a set of dimensional eigenvectors of k+7, and k is the dimension of SURF-BOW. Then the eigenvalues of the key frames composed of a sign language S containing n_n key gestures are n dimensional eigenvectors of k+7. Assuming that the template sign language C contains n_m key frames, the n_n key gesture of the sign language S to be tested and the n_m key gesture of the template sign language C are $S = \{\omega_{S1}, \omega_{S2}, \cdots, \omega_{Sn_n}\}$ and $C = \{\omega_{C1}, \omega_{C2}, \cdots, \omega_{Cn_m}\}$ .The $n_n \times n_m$ matrix D is constructed, and the i-row j-column element d_{ij} of the matrix D represents the Euclidean distance of ω_{Ci} and ω_{Sj} .Then a path can be found in matrix D such that the sum of d through which it passes is minimal. Define the distance between S and C as:

$$Dis(S,C) = \frac{\min\left\{\sum_{(i,j)\in L} d_{ij}\right\}}{\min(n_m, n_n)} \tag{15}$$

The template sign $C_{\min}$ with the shortest distance Dis from the sign language S to be tested is selected as the final decision category of S. When the value of the shortest distance $Dis(S, C_{\min})$ is greater than the error threshold, it is determined that the recognition has failed.

IV. Experiment and Analysis

First, 70 sign language words were selected for the experiment. The 70 sign language words contain 20 one-handed sign language words and 50 two-handed sign language words. The sign language words vary in length and duration is between 1s and 4s. As shown in Table 1.

According to the 70 sign language words, an experimental database was established: a total of 5 sign language players were selected, and the sign language of the above 70 sign language words was collected for each sign language by Kinect2.0. There were 350 sets of sign language data.

TABLE I. Experimental Sign Language

	One-handed vocabulary (20)	Two-hand vocabulary (50)
Vocabulary name	Good, authority, every day, goodbye, doubt, you, me, thought, fantasy, restraint, child, grandmother, mother, us, greeting, high speed, salute, thank you, sky, organization	Labor, grandfather, sack, handkerchief, engagement, hesitation, state, competent, attendance, appraisal, hire, someone, people, recommendation, completion, adaptation, influence, suggestion, communication, correction, study, school, discussion, fly, honest , brave, early leave, quota, belt, vest, socks, around, opposite, self-employed, accept, meet, close, help, guarantee, instruction, criticism, implementation, steps, finishing, jacket, belt, divorce, sun, interaction, Family

TABLE II. COMPARISON OF TRAJECTORY MATCHING AND TRAJECTORY + HAND MATCHING RECOGNITION RATE

Number of experiments	1	2	3	4	5	Average recognition rate
Trajectory matching	82.8%	80%	78.6%	85.7%	82.8%	81.98%
Trajectory + hand match	91.4%	88.5%	85.7%	95.7%	92.9%	90.84%

Five experiments were performed on the data of 50 two-hand sign language in the above experimental database, and the sign language data of one of the sign language players was used as the test sample, and the sign language data of the other four persons was used as the template sample. Two sign language recognition strategies, "only recognize the hand motion trajectory" and "trajectory-based and gesture-assisted" mentioned above, were used respectively, and the accuracy of the two strategies was compared. Only using the hand motion trajectory as the recognition basis is to directly use the template type with the minimum weighted DTW matching distance of the trajectory to be measured as the final recognition result. The trajectory matching in both recognition strategies use the improved DTW algorithm as the recognition algorithm. The results are shown in Table 2.

The time-consuming algorithm is critical for real-time sign language recognition. The traditional DTW algorithm and the improved DTW algorithm were used to match the hand trajectory of the sign language in the experimental database, and the time required for each sign language trajectory was recorded, and the average value of the one-hand sign language and the two-hand sign language was obtained. In order to eliminate the time impact of other factors, this experiment only performed trajectory matching. The results are shown in Table 3.

TABLE III. TIME-CONSUMING COMPARISON OF TRADITIONAL DTW AND IMPROVED DTW

Algorithm	Traditional DTW	Improved DTW
One-handed track average time (s)	0.31	0.24
Two-handed track average time (s)	0.54	0.39

The matching speed using the improved DTW algorithm was significantly higher than traditional. The effectiveness of the proposed DTW algorithm is verified.

As the number of sample templates increases, the time taken to match using the DTW algorithm will increase. Although using endpoint relaxation boundary constraint can't reduce the time complexity of DTW matching, can effectively reduce the amount of computation in the DTW matching process. The partial sequence is eliminated by the lower bound function, which directly reduces the number of template samples. There are more sample symbols in two-hand sign language, and each sign language needs to calculate the left and right time series separately, so the effect of LB_BC lower bound distance filtering is more obvious.

V. CONCLUSION

The paper studies the matching algorithm of dynamic sign language: DTW algorithm. Firstly, the principle of DTW is optimized by endpoint relaxation boundary constraint and early termination matching, and the LB_BC lower bound function is used to filter some of the template to be tested.

Next, the paper introduces the sign language recognition strategy. First, the matching success rate is improved by assigning weights to the bone points according to the magnitude of the motion of the bone points on the hands. A method for extracting key frames in a trajectory density curve using a sliding window is proposed. Then, the feature vector of the gesture of the key frame is extracted, and the sign language recognition is performed by the method of identifying the motion trajectory and identifying the key gesture.

Finally, the paper experiments and analyzes the recognition rate of trajectory matching and key gesture matching, and the time-consuming improvement of DTW, and then verifies the effectiveness of the algorithm and improvement. Experiments show that the proposed method can realize the recognition of dynamic sign language well and has good real-time performance. It has improved the recognition speed and recognition accuracy rate compared with the traditional DTW algorithm.

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